READORA OPEN ACADEMIC TEXTBOOK SERIES

Strength of Materials & Structures

Stress-Strain Mechanics, Torsion & Structural Deflection

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Preface & Curriculum Overview

This textbook has been compiled for undergraduate and postgraduate students enrolled in the **Civil Engineering (CIVIL)** program. It provides structured, rigorous theoretical models, practical case demonstrations, analytical formulas, and chapter-end review exercises. Students are encouraged to practice the core derivations and apply algorithmic workflows to real-world engineering challenges.

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Chapter 1: Fundamental Stress and Strain Relationships

1.1 Normal Stress, Shear Stress and Hooke Law

Internal resisting forces per unit area characterize stress distributions. In the linear elastic regime, stress is directly proportional to strain via Young Modulus: sigma = E * epsilon.

> ***Key Concept:*** *Poisson ratio characterizes lateral contraction perpendicular to applied tensile elongation.*

Review Questions & Laboratory Exercises

- Construct Mohr Circle of Stress for a two-dimensional state of plane stress.
- Calculate principal stresses and maximum in-plane shear stress.

Chapter 2: Bending Stresses & Shear in Beams

2.1 Pure Bending and the Flexure Formula

Transverse loading creates internal bending moments. Normal bending stress varies linearly from the neutral axis: sigma = -(M * y) / I.

> ***Key Concept:*** *Section modulus Z = I/c dictates beam structural resistance against flexural failure.*

Review Questions & Laboratory Exercises

• Draw shear force and bending moment diagrams (SFD & BMD) for a simply supported beam with uniformly distributed load (UDL).

• Derive the elastic curve deflection equation via double integration.

Chapter 3: Torsion of Circular Shafts and Column Buckling

3.1 Torsional Shear Stress in Shafts

Twisting moments produce shearing strain proportional to radial distance from the polar centroid: tau = (T * r) / J.

> ***Key Concept:*** *Euler critical buckling load for slender columns is P_cr = (pi^2 * E * I) / (K * L)^2.*

Review Questions & Laboratory Exercises

- Determine the required diameter of a solid steel transmission shaft under specified torque.
- Explain effective length factors for different column boundary end conditions.